Questions	Answers	Topic
Q01: Can we repurpose sensor/devices our other collaborators have developed (instead of developing something new)?	A01: Yes, as long as the other devices are still developmental prototypes.	CBA-01
Q02: Do you have interest in leveraging modeling and machine-learning-based approaches as part of the sensor platform prototype?	A02: Yes	CBA-01
Q03: Is 10 ppb across threat space an appropriate LOD?	A03: At this point in development we aren't as concerned about LOD, rather the detector's ability to measure within a material/substrate. We will establish LODs once we understand feasibility. That being said, being more sensitive is always better.	CBA-01
Q04: Are there specifications for the expected mass per unit volume of CWAs absorbed into the surface.	A04: No	CBA-01
Q05: Does the detector have to be on-the-move while collecting data? If so are there minimum vehicle speeds that have to be achieved?	A05: No	CBA-01
Q06: Can the senor/sensors come into contact with the surface? If the sensors are not able to contact the surface is there a minimum standoff from the sensor to the surface?	A06: Direct contact with the surface is acceptable.	CBA-01
Q07: Is there a required time (quantified in seconds) to detect? (Stated another way, is there an interrogation scan area/unit time requirement?)	A07: Time to detect is not a critical measure for this technology, however a higher throughput could provide additional operational benefit.	CBA-01
Q08: Is there a particle size or range of sizes for the bioaerosols that are of particular interest?	A08: 1- 10 micron would be the appropriate size range.	CBA-02
Q09: Is the intention for this proposal to be integrated into an immune-building scenario for indoor testing?	A09: There is no pre-conceived intention. We ask offerors to suggest appropriate answers.	CBA-02
Q10: Will the prototype instrument be required to contain wind and environmental-tracking instruments itself, or is the intention for the instrument to be integrated with external instruments?	A10: There is no requirement for on-board wind and environmental tracking instruments. It is likely the instrument will be synchronized with external instrumentation.	CBA-02
Q11: For the autonomous monitoring portion, how often will samples need to be collected and analyzed (every half-hour, hour, day, etc.)? Is there a preference for continuous monitoring vs. collect/detect only after an anomalous event?	A11: Offerors should suggest sampling interval(s). There is a preference for sample collection only after an anomalous event.	CBA-02
Q12: How long will the instrument need to be able to run autonomously (days, weeks, etc.)?	A12: Offerors should suggest interval(s). This will typically be dependent on reagent usage and storage conditions.	CBA-02
Q13: There are some portable systems currently on the market that include several components listed for bioaerosol detection, such as portable MALDI-MS. Are there any technologies that are not of interest for this call?	A13: DTRA is interested in all technologies that are applicable to the call.	CBA-02
Q14: Is this BAA topic projected to be tied to any future advanced development efforts?	A14: No, this is an initial project to develop a combined sensor/sampler/analyzer/reporter. When it works further development is bound to continue but at this point we are not tied to any specific effort.	CBA-02
Q15: Is the intent of CBA-03 to develop a framework platform and modular plug-in-play tool/algorithm options to enable others (e.g., DOD laboratory end-users) to customize data processing , or is the intent of CBA-03 for the awardee to develop and then implement the algorithm(s) to answer questions about samples (characterize, predict, etc.) for DoD laboratory end-users?	A15: The CBA-03 is seeking innovation solutions for both: Development of modular, plug-in-play algorithms, models and relevant workflows specifically for curation, harmonization and integration of data and results from across types of omics to improve the accuracy and timeliness of detection of biothreats, prediction of emerging and unknown threats, increase the in-depth understanding, and shorten the threat analysis time; And development of a robust and scalable computing infrastructure/platform capable to host the modular plug-in-play tools/algorithms, and compatible with cloud based and on-premise environments . An ideal proposal should address both, however solutions for each are also considered.	CBA-03
Q16: For CBA-03, is the intent that the performer will be analyzing data sets (generated on project and/or open-source) to benchmark models, tools and data workflows?	A16: The offeror will be analyzing data sets to develop algorithms, models and workflows using raw data sets. Please note that the raw data should be analyzed (pre-processed) using user's choice software (i.e. commercial software) which is not hosted in the seeking platform. The pre-processed data will be uploaded the SMOIS-PAD for further multi-omic analysis.	CBA-03

Questions	Answers	Topic
Q17: For CBA-03, in reference to cloud computing, what environment is preferred (AWS, Azure, or both)?	A17: Ideally both.	CBA-03
Q18: For CBA-03, is there a preferred hardware guideline for on-premises environments? And what type of operating system is preferred (e.g., Linux – Red Hat or Ubuntu), or is the intent to integrate against multiple OS types (and if so, which are preferred)?	A18: Applications should support deployment on Linux_x86, Linux_ARM, and Windows enabled devices.	CBA-03
Q19: For CBA-03, is containerization preferred and if so, is there a preferred option (e.g., Docker/Singularity, Virtual Machines)?	A19: Yes, containerization is a request, and Docker is preferred.	CBA-03
Q20: For implementing CBA-03, is the data classification level anticipated to be unclassified, controlled unclassified, or classified (or a combination)? And is this anticipated to be the same for both cloud and on-premises instances?	A20: The framework platform, integrated analysis tools, and operational environments are anticipated to be compatible for unclassified, CUI data. Classified data option is preferrable.	CBA-03
Q21: During CBA-03 solutioning, is it anticipated that data and the code be handled at the same or separate classification levels (e.g., code at unclassified, data at classified)?	A21: The code is anticipated to be CUI, and the classification level of omic data could be unclassified, CUI or classified.	CBA-03
Q22: For CBA-03, is compatibility across multiple software development languages expected? If so, can you provide a list?	A22: Anticipated framework platform enables scalable and reproducible scientific workflows using software containers. It allows the adaptation of pipelines written in the most common scripting languages. Selected software languages must support deployment on Linux_x86, Linux_ARM, and Windows enabled devices.	CBA-03
Q23: In the CBA-03 topic call, it states: 'pre-processed using user-choice, no-hosted data processing software. By 'no-hosted,' does this mean non-cloud/locally installed? If not, can you clarify the requirements for something to be 'no-hosted?'	A23: Raw data generated from lab instruments for a type of omics will be pre-processed with a suitable, commercial or open-source data processing software (e.g. Proteome Discoverer for proteomics data collected with Thermo-Scientific Mass Spectrometer). The pre-processing software is a user-choice, and not anticipated to be provided, integrated or hosted in the SMOIS-PAD platform. The pre-processed outcomes are being subjected to further analysis across multi-omics using tools on the SMOIS-PAD.	CBA-03
Q24: For CBA-03, is the intention for the performer to use AI/ML/DL for data standardization primarily, or for analysis as well?	A24: The CBA-03 is seeking solutions for integration and analysis of data across multi-omics rather than individual ones. Data standardization is desired to support data integration.	CBA-03
Q25: Given that the scope of the BAA is limited to projects that meet TRL definitions in the range of 3-6, is it intended that all initial components of the workflow should meet TRL3? If this is the case, is novel AI/ML/DL in scope?	A25: Novel AI/ML/DL solutions are highly recommended. The solutions are anticipated to be validated, fully functioning, and deployable at completion.	CBA-03
Q26: For CBA-03, is the desired end-state a command-line interface, or is a graphical user interface required?	A26: There is no specific requirement, however GUI is highly preferrable.	CBA-03
Q27: For CBA-03, are there preferred (or pre-existing) security solutions and measures already identified that the solution will need to integrate with?	A27: All software deliverables should be in compliance with Iron Bank's Acceptance Baseline Criteria (ABC), which formally documents the hardening and contribution requirements for container images (https://p1.dso.mil/services/iron-bank).	CBA-03
Q28: Do you already have your performers selected for CBA-03, the development of SMOIS-PAD (strategies for multi-omics integration solutions for pathogen-agnostic detection), or are you open to new performers? Some colleagues and I are developing AI systems for precisely these purposes. We have some neat ideas to submit to this call, provided we're able to be considered at this stage.	A28: The SMOIS-PAD topic call is open to the public.	CBA-03
Q29: What is the anticipated level of purity and type of pre-processing envisioned for the samples to be analyzed with this tool?	A29: Software tools and/or solutions proposed should work for samples at any level of purity, ranging from simple to highly complex ones. Raw data generated from a sample analysis will be pre-processed with a suitable, commercial or open-source data processing software (e.g. Proteome Discoverer for proteomics data collected with Thermo-Scientific Mass Spectrometer).	CBA-03
Q30: Is CBA-03 intended to be capable of interrogating samples acquired by the device described in CBA-02?	A30: Yes, it is if possible. However, CBA-03 is not limited for bioaerosols, but broadly capable to being applied for varieties of sample type with high complexity in environmental and medical matrices.	CBA-03
Q31: Is data storage a critical consideration? If so, what amount of storage is needed?	A31: Offerors must provide access to storage capabilities to handle the raw data and metadata generated during this effort and outline those in the technical approach.	CBA-03
Q32: The solicitation describes a scenario where sequencing is done in the field. Are there any power supply constraints, or should the full range of power availability in the field be considered?	A32: The full range of power availability in the field should be considered.	CBA-03

Questions	Answers	Topic
Q33: Will you consider proposals focused on developing multi-omics sensing with electronic devices?	A33: No	CBA-03
Q34: Will you consider the development of low-cost and high-throughput screening tools for the detection of pathogens?	A34: : Yes, we are seeking data science based technical solutions for detection, prediction, and understanding biothreats including pathogens. Low-cost and high-throughput screening solutions (software tools) are considered. We do not call for development of instruments or physical tools for the purposes.	CBA-03
Q35: Will you consider the development of personalized test kits for pathogen detection?	A35: No	CBA-03
Q36: Are there any requirements or options for the timeline such as base and optional or split timelines?	A36: Offerors should propose base and option periods as appropriate.	CBA-03
Q37: We understand from the Topic that the data will be stored on a government platform, but where will the modularized "plug-and-play" applications within the toolkit be stored?	A37: Modules may be stored withing one domain with options of on-premise (one hardware system with periodic updates) or stored on Cloud (i.e. AWS Govcloud with automatic updates) and accessed on demand. Plug-and-play applications should support deployment on Linux_x86, Linux_ARM, and Windows enabled devices.	CBA-03
Q38: Is there additional detail regarding the end-user for the solution beyond DoD laboratories? For example, will it be data engineers within those laboratories or research scientists? We are asking to understand the background of the anticipated end user.	A38: The anticipated end-users will be Warfighters or technical persons that are minimally on the software.	CBA-03
Q39: There is a requirement to "leverage the relevant pre-existing biological knowledge." Can you elaborate on what is meant by pre-existing knowledge? Is this knowledge distinct from pre-existing data?	A39: Capability to leverage biological knowledge databases (i.e. International Taxonomy of Viruses, NCBI, KEGG, Uniprot) and extract datasets from previous/future research projects.	CBA-03
Q40: Are you able to explain the difference between the terms pre-processing and processing? For example, is the processing software mentioned in the market research requirement different from the " the most popular industrial or open-source data pre-processing software"?	A40: The term of Pre-processing implies steps of handling raw data generated from a lab instrument for a type of omics with a suitable, commercial or open-source data processing software (e.g. Proteome Discoverer for proteomics data collected with Thermo-Scientific Mass Spectrometer). The pre-processing software is a user-choice, and not anticipated to be provided, integrated or hosted in the SMOIS-PAD platform. The pre-processed data (e.g. compound IDs, MS intensities, pathways, etc.) are then being processed to further analysis across multi-omics using tools on the SMOIS-PAD.	CBA-03
Q41: To what extent do we need to do harmonization for the data types listed (e.g., pdf, ppts, hardware…)? For example, are we extracting metadata from pdfs? Data from images?	A41: Submissions must address the generalizability of the proposed solution to extract, harmonize, and correlate different multi-omic data sets across heterogeneous data types and reporting formats (e.g., Word documents, PDFs, PPTs, Hardware (hard drives), Instrument data files, etc.).	CBA-03
Q42: To clarify, does trustworthiness/explainability for data refer to quality control of said data? Is model trustworthiness/explainability required as well?	A42: Yes, Data must be accessible, secure, traceable, privacy protected, accurate and AI ready. Offerors should refer to the DOD Data, Analytics, and Artificial Intelligence Adoption Strategy.	CBA-03
Q43: Is there an expected time frame for the skin/oral MPS to be viable (i.e. days, weeks, months)?	A43: The intent is to exploit changes in the microbiome that could be used for diagnostics, therefore, the model system should be able to track changes post exposure within days. The MPS should be viable long enough to obtain clear data about normal baseline markers as well as to collect data pre- and post-exposure to agents that is biologically relevant.	CBA-04
Q44: What are top priorities for target agents for this call topic?	A44: The top priorities for target agents would be highly contagious viruses.	CBA-04
Q45: Which biological agents are the greatest risks to the warfighter, that may impact the skin or oral microbiome?	A45: The Offeror should propose the appropriate agent(s), detail why that considered agent(s) is/are a risk, and why the skin/oral microbiome MPS is a good system to study the agent(s) in. Viral agents are the highest priority for this topic.	CBA-04
Q46: Is the machine learning element solely focused on streamlining analysis from future clinical samples or does CBA Medical Diagnostics desire any type of predictive modeling capability from the ML component?	A46: The focus of this topic is to develop a model system to detect changes in the microbiome that can be exploited for diagnostics. Systems that incorporate predictive modeling will be considered if they address the original intent.	CBA-04

Questions	Answers	Topic
Q47: Are the multi-omics targets intended to be a fixed panel for analysis or discovered via sequencing?	A47: The multi omic targets would be determined using the model system and comparing healthy state to disease state. Please keep in mind that using a fixed panel for analysis may limit and miss potential novel markers of pre-symptomatic exposure.	CBA-04
Q48: Is there interest in an affordable multi-omic end-point analysis with fixed biomarker targets?	A48: The model system will be used to identify which biomarker targets are of interest.	CBA-04
Q49: What is an ideal number of multi-omic targets?	A49: This can not be determined without first determining the effect of infection on the healthy microbiome. The number of multi-omic targets is at the discretion of the performer but should be enough to paint a clear picture of baseline oral/skin microbiome MPS markers as well as to be able to examine the marker landscape post-exposure to a biological agent(s).	CBA-04
Q50: What is the ideal cost per run for the multi-omic analysis?	A50: The cost per run is at the discretion of the performer. The focus of this topic should be on creating a model system that can identify the proper markers for future diagnostic development, it is not intended to develop a diagnostic.	CBA-04
Q51: Are you interested in analyzing the cellular immune response when there is an imbalance in human tissue or microbiome in the presence of biological agents?	A51: No, the intent is to determine whether changes in the microbiome can be used as a diagnostic for early stages of infection.	CBA-04
Q52: What types of specific endpoint assays/functional assays are you interested in? For the skin cellular component, cell barrier integrity, cell viability, inflammatory response, and metabolic activity; for bacteria, specific microbiome composition and fold changes.	A52: This is up to the discretion of the offerors.	CBA-04
Q53: Do you support the development of personalized MPS models?	A53: No	CBA-04
Q54: Do you support the study of integrating MPS and -omics analysis, and which types of -omics analysis are they most interested in?	A54: Yes. We are interested in omic analysis that could characterize shifts in the microbiome. We support studying integration of -omics analysis and MPS.	CBA-04
Q55: Are we limited to ESKAPE pathogens, or can we also look at fungal infections?	A55: Viral agents are the highest priority for this topic. However, the offeror is not limited to any set of pathogens, but keep in mind that the oral/skin microbiome should be representative of the normal human flora.	CBA-04
Q56: Is there a preference for what type (cell type) of skin model?	A56: No	CBA-04
Q57: Are we limited to FDA-approved antibacterial or antifungal compounds?	A57: The focus of the topic is not antibacterial or antifungal compounds. We are interested in how infections cause changes in the microbiome and whether such changes could be detected and exploited for diagnostic purposes.	CBA-04
Q58: For multiomic analysis, are we restricted to mammalian biomarkers, or can we also look at microbial biomarkers?	A58: Microbial biomarkers should be the primary focus, but changes in the human biomarker landscape would also be informative for the purposes of diagnostic development.	CBA-04
Q59: Can you expand on the biological agents of interest relevant to the RD-CB portfolio or within the context of the skin and oral microbiome MPS models?	A59: The Offeror should propose the appropriate agent(s), detail why that considered agent(s) is/are a risk, and why the skin/oral microbiome MPS is a good system to study the agent(s) in. Viral agents are the highest priority for this topic.	CBA-04
Q60: Are other CBRN exposures within scope? Specifically, are chemical or radiological exposures of interest?	A60: No	CBA-04
Q61: Are micro-physiological systems (MPS) that are not microfluidic platforms within scope?	A61: Yes	CBA-04
Q62: What is the vision for a biological threat/microbiome impact? One version of this focuses on threats that would specifically be expected to encounter the skin/oral microbiomes (topical exposures), while another could view this as systemic exposure/infection that activates the host immune system and secondarily affects the topical microbiomes. Can you share DTRA's vision for the topic of primary vs. secondary effects on topical microbiomes?	A62: The vision for this effort is to see if we can exploit the microbiome for diagnostic purposes. In other words, if an individual is infected with an agent that leads to a systemic infection, does initial contact with that agent cause changes in the oral or skin microbiome that could indicate that they are in the early stages of infection. Secondary to this, we are interested in local and systemic changes in the host that could have an appreciable effect on the microbiome that can be detected.	CBA-04

Questions	Answers	Topic
Q63: Can DTRA share more about biological threats of interest to the agency?	A63: The Offeror should propose the appropriate agent(s), detail why that considered agent(s) is/are a risk, and why the skin/oral microbiome MPS is a good system to study the agent(s) in. Viral agents are the highest priority for this topic.	CBA-04
Q64: What is the timeline post-threat exposure that is most of interest for "pre-symptomatic indicators"? What about timeline post sample-collection?	A64: The timeline for pre-symptomatic should be from the moment of exposure up to when symptoms are expected from normal human exposure to an agent. We are unsure what "post sample-collection" is referring to in this question. However, we would be interested in data collected outside of the pre-symptomatic window.	CBA-04
Q65: Does CBA Medical Diagnostics have a preference between a singular representative microbiome MPS for the skin and/or oral cavity (i.e., a core microbiome) versus a modifiable representative microbiome MPS that could be tailored and/or tuned to represent microbiomes from different geographical regions, genders, races, environmental factors, etc. (i.e., a flexible core microbiome)?	A65: Yes, the microbiome used in this initial effort should consist of the core microbiome for the broadest applicability. Later iterations will likely address the flexible microbiome. It is important to keep in mind that this effort aims to benefit the warfighter.	CBA-04
Q66: Is the scope for this BAA focused on a healthy microbiome, or would any diseased state microbiomes also fall within scope?	A66: The purpose is to see if a shift from a "healthy" microbiome to disease state can be modeled and detected.	CBA-04
Q67: What is the budget limit, or recommended budget for applications to the BAA?	A67: The offeror should submit what they deem appropriate for their submission to the topic(s).	General
Q68: Are there any requirements or definitions for the allocation of budget for collaborators?	A68: The offeror should submit what they deem appropriate for their submission to the topic(s).	General
Q69: For CBA-03 Is there a preferred font, and font size, for the Quad Chart and White Paper?	A69: No	General
Q70: Required submission through www.dtrasubmission.net. We've tried to access this link, but it has an error and not found. This is not the first time we've had this issue. Please advise how to proceed.	A70: The site was momentarily down but is working again. If the issue persists, contact the help desk immediately: help@dtrasubmission.net.	General
Q71: Are there any requirements (or limitations) for employee citizenship?	A71: Employment of Foreign Nationals must be reported to DTRA for appropriate review and instruction in compliance with Security Classification Guidance.	General
Q72: Will multiple awards be given for this opportunity?	A72: DTRA may award one or multiple awards.	General
Q73: Is there an expected period of performance for the effort (e.g. 12 months, 24 months etc.)?	A73: The offeror should submit what they deem appropriate for their submission to the topic(s).	General
Q74: Please confirm this Topic follows the two phase proposal process described in the BAA. Hence, submissions due by May 29 follow the instructions provided in Section 4.6.1 of the BAA.	A74: This call will use a two-phased evaluation process with Phase I proposals, as described on SAM.gov, due 5/29/24. If selected to submit a full proposal, the Phase II due date will be announced in the e-mail sent to the performer.	General
Q75: Can the government provide a prospective award date for planning purposes?	A75: Estimated Award Date is on or about the 2nd quarter 2025.	General
Q76: Assuming viable proposals, how many awards is the government planning for this topic?	A76: DTRA may award one or multiple awards.	General
Q77: Will multiple awards be given for this opportunity?	A77: DTRA may award one or multiple awards.	General
Q78: Will this be a 2 phase process for submission? Does that mean Phase 1 only the quad chart, white paper, and SOW due on May 29? What will be the Phase 2 due date?	A:78 This call will use a two-phased evaluation process with Phase I proposals, as described on SAM.gov, due 5/29/24. If selected to submit a full proposal, the Phase II due date will be announced in the e-mail sent to the performer.	General
Q79: Can you confirm that only Phase 1 components are due 5/29/24.	A:79 This call will use a two-phased evaluation process with Phase I proposals, as described on SAM.gov, due 5/29/24. If selected to submit a full proposal, the Phase II due date will be announced in the e-mail sent to the performer.	General

Questions	Answers	Topic
Q80: Is CBA-03 a one-phase or two-phase submission process? If two-phase, will both submissions for Phase-1 and Phase-2 occur before May 29th?	A:80 This call will use a two-phased evaluation process with Phase I proposals, as described on SAM.gov, due 5/29/24. If selected to submit a full proposal, the Phase II due date will be announced in the e-mail sent to the performer.	General
Q81: What is the estimated award value for this BAA/topic 4?	A81: The offeror should submit what they deem appropriate for their submission to the topic(s).	General
Q82: What is the period of performance?	A82: The offeror should submit what they deem appropriate for their submission to the topic(s).	General